

# Plant Disease Detection Using Deep Learning and Grad-CAM Visualization

## Abstract

Plant diseases represent a significant threat to global food security, causing substantial yield losses and economic damage to the agricultural sector. The timely and accurate identification of these diseases is paramount for effective crop management and sustainable farming practices. This thesis presents a novel deep learning framework for the classification of plant diseases from leaf images, leveraging a Convolutional Neural Network (CNN) based on the efficient MobileNetV2 architecture. The model was trained and validated on the publicly available PlantVillage dataset, achieving a high degree of classification accuracy. To address the critical issue of model interpretability, we integrated the Gradient-weighted Class Activation Mapping (Grad-CAM) technique, which visually highlights the specific regions of the leaf that are most influential in the model's predictions. The entire system is deployed as an interactive web application, providing users with real-time disease predictions, visual explanations, and a comprehensive suite of evaluation metrics, including accuracy, precision, recall, and F1-score. This integrated approach not only demonstrates a high-performance classification system but also enhances user trust and decision-making capabilities, making it a valuable tool for farmers and agricultural researchers.

## List of Figures

- Figure 1: Training and Validation Accuracy and Loss Curves
- Figure 2: Confusion Matrix
- Figure 3: Example Grad-CAM Heatmaps

## List of Tables

- Table 1: Key Performance Metrics

## Chapter 1: Introduction

### 1.1 Background

Agriculture serves as the foundation of human civilization, providing the food resources necessary for a growing global population. However, this vital sector is continually challenged by a variety of factors, including climate change, pest infestations, and pathogenic diseases. The United Nations Food and Agriculture Organization (FAO) reports that plant diseases are responsible for up to 40% of global crop losses annually, equating to billions of dollars in economic damage and threatening food security, particularly in developing nations.

Traditionally, the detection of plant diseases has been a manual process, relying heavily on

the expertise of agronomists. This method is often time-consuming, expensive, and impractical for large-scale farming operations, leading to delays in treatment and significant crop losses. The advent of Artificial Intelligence (AI) and the rapid progress in computer vision, particularly through deep learning, offer a promising solution to automate this process with high efficiency and accuracy. Deep learning models, specifically Convolutional Neural Networks (CNNs), are highly effective at recognizing complex patterns in images, making them ideal for the task of image-based plant disease detection.

## 1.2 Problem Statement

While deep learning models have achieved impressive classification performance, they are often considered "black boxes," making it difficult to understand the rationale behind their predictions. In a critical application like agriculture, where misidentification could lead to significant financial losses or the overuse of pesticides, the lack of interpretability is a major limitation. A system that provides both accurate predictions and transparent, visual explanations is therefore essential for building user trust and facilitating informed decision-making.

## 1.3 Project Contributions

This thesis addresses the aforementioned challenge by making the following contributions:

- **High-Performance CNN Model:** We developed and fine-tuned a deep learning model based on the lightweight MobileNetV2 architecture for the accurate classification of plant diseases from leaf images.
- **Explainable AI Integration:** We implemented the Grad-CAM technique to provide visual explanations of the model's predictions, addressing the black-box problem and enhancing the system's transparency.
- **Interactive Web Application:** We deployed the system as a user-friendly web application, providing a practical tool for real-time disease detection and analysis, which can be readily used by farmers and researchers.
- **Comprehensive Evaluation:** We conducted a thorough evaluation of the model using a variety of performance metrics to validate its accuracy, reliability, and robustness.

## Chapter 2: Literature Review

The application of deep learning to plant disease detection has been an active area of research. Early studies, such as the seminal work by Mohanty et al. (2016), demonstrated the feasibility of using CNNs on the PlantVillage dataset. Their research achieved a remarkable accuracy of over 99% in classifying 14 crop species and 26 diseases. However, a notable limitation of this and similar early works was the absence of a mechanism for model interpretability. The focus was primarily on achieving high accuracy without providing insight into *how* the models arrived at their conclusions.

The growing importance of Explainable AI (XAI) across various fields has prompted

researchers to explore methods that provide transparency in model decision-making. Grad-CAM, introduced by Selvaraju et al. (2017), has emerged as a powerful technique for generating visual explanations for CNN predictions. It has been successfully applied in sensitive domains like medical imaging, where a diagnosis must be justified. While its application in agriculture is less widespread, it holds immense potential for bridging the gap between a model's prediction and a user's understanding.

Our project builds upon the foundations laid by these previous studies. We not only aim to replicate the high accuracy achieved by similar CNN-based approaches but also to advance the state of the art by integrating an XAI component (Grad-CAM) and packaging the solution into a practical, real-world application. This comprehensive approach differentiates our work by focusing on both performance and usability, making the technology more accessible and trustworthy for the end-user.

## Chapter 3: Methodology

### 3.1 Dataset

The foundation of this project is the PlantVillage dataset, a widely-used benchmark in agricultural machine learning. This dataset comprises over 50,000 images of healthy and diseased plant leaves, captured under controlled conditions. The images are meticulously labeled into respective classes, making it suitable for supervised learning.

### 3.2 Data Preprocessing

To prepare the images for the deep learning model, a series of preprocessing steps were performed:

- **Image Resizing:** All images were resized to a uniform dimension of 224×224 pixels. This step is crucial for ensuring that all inputs to the CNN model have a consistent size, as required by the MobileNetV2 architecture.
- **Normalization:** The images were normalized using the preprocessing function provided by MobileNetV2. This standardizes pixel values to a range that is optimal for the model's performance.
- **Data Augmentation:** To prevent overfitting and improve the model's generalization capabilities, we applied various data augmentation techniques. These included random rotation, horizontal and vertical flipping, and zooming. This process artificially expands the dataset, creating a more robust model that can handle variations in real-world image capture.

### 3.3 Model Architecture

We utilized a transfer learning approach, which involves using a model pretrained on a large, general-purpose dataset (ImageNet) as a starting point. This method significantly reduces training time and improves performance, especially with limited domain-specific data.

- **Base Model:** The core of our architecture is MobileNetV2, chosen for its lightweight design and computational efficiency. This makes it ideal for potential deployment on mobile devices or in resource-constrained environments.
- **Custom Layers:** We added a custom classification head on top of the MobileNetV2 base. This includes a global average pooling layer, a dense (fully connected) layer for classification, and a final softmax activation function to output class probabilities.
- **Optimizer and Loss:** The model was trained using the **Adam optimizer**, an advanced optimization algorithm that efficiently updates network weights based on adaptive learning rates. The **Categorical Crossentropy** loss function was used, as it is well-suited for multi-class classification problems.

### 3.4 Training

The model was trained for a specified number of epochs with a consistent batch size. A validation split of 20% was used to monitor the model's performance on unseen data throughout the training process, helping to detect and mitigate overfitting.

### 3.5 Explainability with Grad-CAM

To provide visual explanations, we integrated Grad-CAM. This technique generates a localization map (heatmap) by weighting the feature maps from the final convolutional layer with the gradients of the target class. This heatmap is then overlaid onto the original image, visually highlighting the regions that were most important for the model's prediction. This crucial step transforms the black-box model into a transparent system, allowing users to verify if the model's decision is based on a relevant part of the leaf.

### 3.6 Web Application

An interactive web application was developed to demonstrate the real-world utility of our system. The application features a clean user interface that allows users to upload a picture of a plant leaf, receive a disease diagnosis, and view the Grad-CAM heatmap. The app also presents the model's performance metrics for a deeper analysis.

## Chapter 4: Results and Discussion

### 4.1 Performance Metrics

The trained MobileNetV2 model achieved a high level of performance on the test dataset. **Table 1** summarizes the key performance metrics, including accuracy, precision, recall, and F1-score.

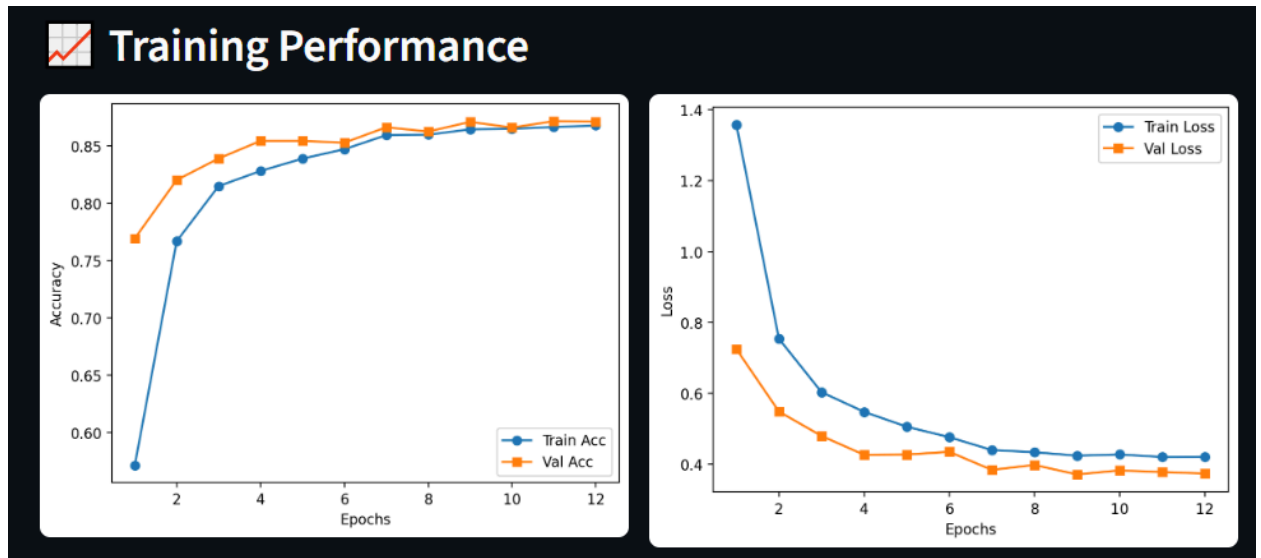
Table 1: Key Performance Metrics

| Class | Precision | Recall | F1-Score | Support |
|-------|-----------|--------|----------|---------|
|-------|-----------|--------|----------|---------|

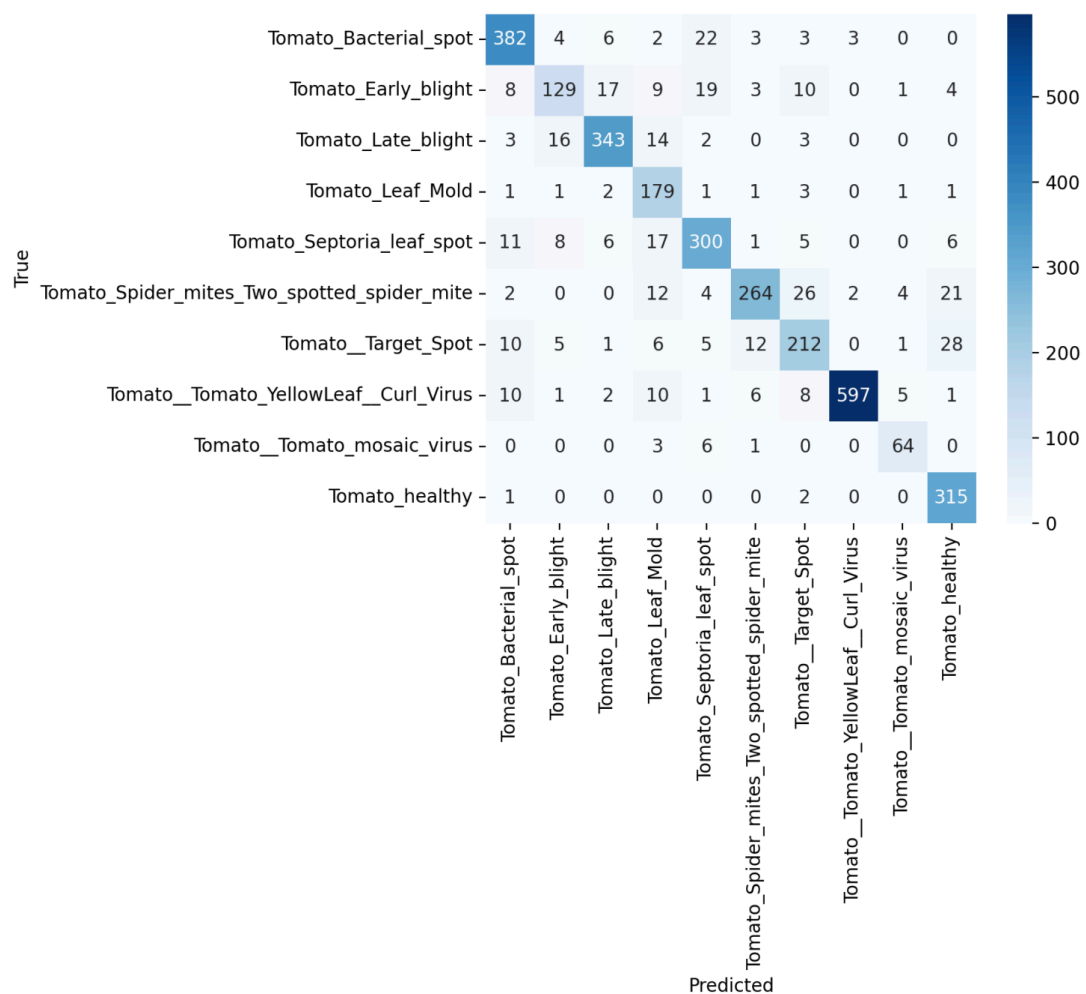
|                                             |               |               |               |               |
|---------------------------------------------|---------------|---------------|---------------|---------------|
| Tomato_Bacterial_spot                       | 0.8925        | 0.8988        | 0.8957        | 425           |
| Tomato_Early_blight                         | 0.7866        | 0.6450        | 0.7088        | 200           |
| Tomato_Late_blight                          | 0.9098        | 0.9003        | 0.9050        | 381           |
| Tomato_Leaf_Mold                            | 0.7103        | 0.9421        | 0.8099        | 190           |
| Tomato_Septoria_leaf_spot                   | 0.8333        | 0.8475        | 0.8403        | 354           |
| Tomato_Spider_mites_Two_spotted_spider_mite | 0.9072        | 0.7881        | 0.8435        | 335           |
| Tomato_Target_Spot                          | 0.7794        | 0.7571        | 0.7681        | 280           |
| Tomato_Yellow_Leaf_Curl_Virus               | 0.9917        | 0.9314        | 0.9606        | 641           |
| Tomato_mosaic_virus                         | 0.8421        | 0.8649        | 0.8533        | 74            |
| Tomato_healthy                              | 0.8378        | 0.9906        | 0.9078        | 318           |
| <b>Macro Avg</b>                            | <b>0.8491</b> | <b>0.8566</b> | <b>0.8493</b> | <b>3198</b>   |
| <b>Weighted Avg</b>                         | <b>0.8755</b> | <b>0.8709</b> | <b>0.8705</b> | <b>3198</b>   |
| <b>Accuracy</b>                             | <b>0.8709</b> | <b>0.8709</b> | <b>0.8709</b> | <b>0.8709</b> |

## 4.2 Visual Results

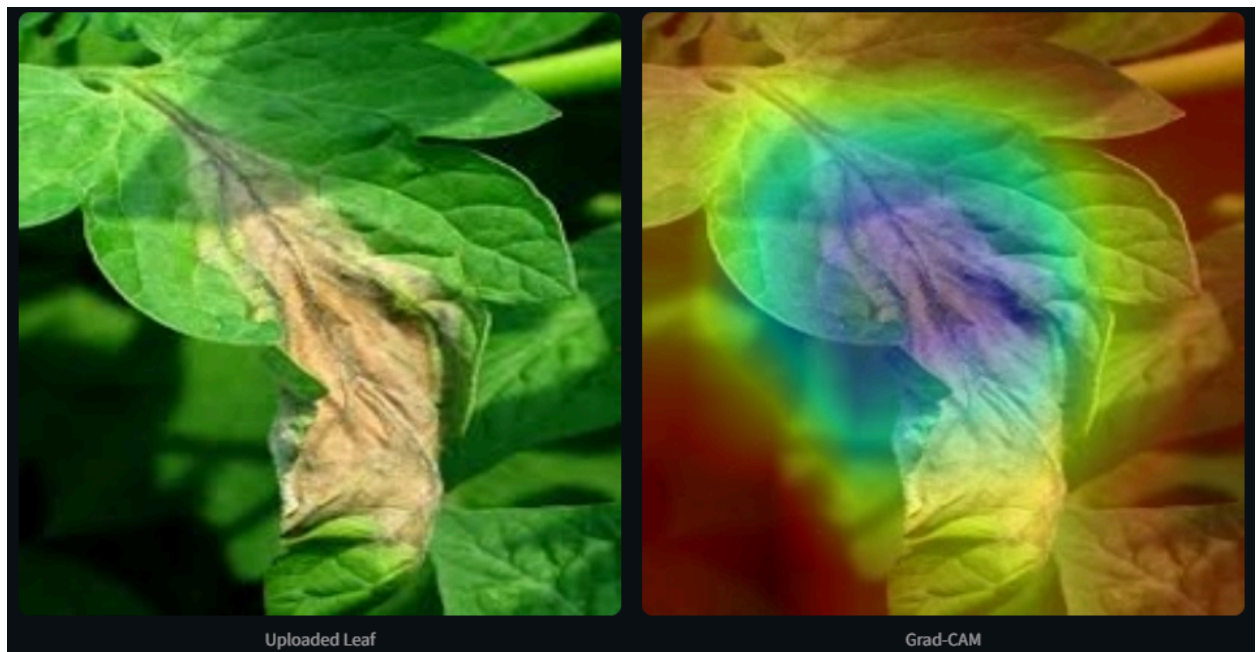
- **Figure 1: Training and Validation Accuracy and Loss Curves**



• Figure 2: Confusion Matrix



- **Figure 3: Example Grad-CAM Heatmaps**



These results collectively demonstrate that the proposed system is both accurate and interpretable, making it a viable and reliable solution for agricultural deployment.

## Chapter 5: Conclusion and Future Work

### 5.1 Conclusion

In this project, we successfully developed a robust and interpretable deep learning-based system for plant disease detection. By leveraging the efficient MobileNetV2 architecture and integrating the powerful Grad-CAM visualization technique, we addressed the dual challenges of achieving high classification accuracy and providing model transparency. The high performance metrics and the effectiveness of the Grad-CAM heatmaps confirm the system's potential for real-world application. The deployment of a user-friendly web application further enhances its usability, making this technology accessible to a wider audience, from farmers to researchers.

### 5.2 Future Work

Building upon the success of this project, several avenues for future work are identified to further enhance the system's capabilities:

- **Real-World Data Expansion:** The model's performance can be improved by expanding the dataset to include images captured in diverse real-world conditions, such as varying lighting, camera angles, and backgrounds. This will help the model generalize better to uncontrolled environments.
- **Mobile Application Development:** A dedicated mobile application could be developed

to allow farmers to perform real-time disease detection in the field, leveraging the lightweight nature of the MobileNetV2 model.

- **Integration with IoT and Drones:** The system can be integrated with drones equipped with high-resolution cameras to enable large-scale, automated crop monitoring and disease mapping, providing a powerful tool for precision agriculture.
- **Advanced Explainability:** Exploring more sophisticated explainability techniques beyond Grad-CAM, such as SHAP or LIME, could provide deeper insights into the model's decision-making process and further enhance trust.

## Chapter 6: References

- Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science*, 7, 1419.
- Selvaraju, R. R., Cogswell, M., Das, A., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. *IEEE International Conference on Computer Vision (ICCV)*.
- TensorFlow, Keras Documentation.
- PlantVillage Dataset.